Model GS-700
Imaging Densitometer

Instruction Manual

Catalog Numbers
170-7601
170-7602

For Technical Service
Call Your Local Bio-Rad Office or
in the U.S. Call **1-800-4BIORAD**
(1-800-424-6723)

PN 4000062 RevB

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Warranty Statement

Bio-Rad Model GS-700 Imaging Densitometer

This warranty may vary outside of the United States. Please contact your local Bio-Rad office for the exact terms of your warranty.

Bio-Rad Laboratories warrants to the customer that the Model GS-700 Imaging Densitometer (Catalog Numbers 170-7601, 170-7602) will be free from defects in material and workmanship, and will meet all performance specifications for the period of one year from the date of shipment. This warranty covers all parts and labor.

In the event that the instrument must be returned to the factory for repair under warranty, the instrument must be packed for return in the original packaging.

Bio-Rad shall not be liable for any incidental, special or consequential loss, damage, or expense directly or indirectly arising from the use of the Model GS-700 Imaging Densitometer. Bio-Rad makes no warranty whatsoever in regard to products or parts furnished by third parties, such being subject to the warranty of their respective manufacturers. Service under this warranty shall be requested by contacting the customer's nearest Bio-Rad office.

This warranty does not extend to any instruments or parts thereof that have been subject to misuse, neglect or accident, or that have been modified by anyone other than Bio-Rad or Bio-Rad's authorized agent, or that have been used in violation of Bio-Rad instructions.

For any inquiry or request for repair service, contact Bio-Rad Laboratories. Inform Bio-Rad of the model and serial number of your instrument.

This Bio-Rad instrument is designed and certified to meet EN 60950, EN 55022-Class B, and EN 55014-Class B safety standards. Certified products are safe to use when operated in accordance with the instruction manual. This instrument should not be modified or altered in any way.

Caution: This equipment generates, uses, and can radiate radio frequency energy and, if not installed in accordance with the instructions manual, may cause interference to radio communications. It has been tested and found to comply with the limits for a Class A computing device pursuant to Subpart J of Part 15 of FCC rules, which are designed to provide reasonable protection against such interference when operated in a commercial environment. Operation of this equipment in a residential area is likely to cause interference. Under these circumstances, the user, at his/her own expense, will be required to take whatever measures may be required to correct the interference.

Section 1
General Information

1.1 Safety

The Model GS-700 Imaging Densitometer system uses high voltage and current and should be operated with care at all times. To avoid shock, set up the GS-700 Densitometer in a dry area. Immediately wipe up any spilled solutions.

- Turn off the densitometer, and then disconnect the power plug when you want to clean the densitometer case or glass plate, or when the densitometer needs service or repair.
- Place the densitometer on a level surface.
- To ensure proper ventilation, allow a minimum of 15 cm free space around each side of the densitometer.
- Do not leave photographs, gels, or film on the glass plate for excessive periods of time. The heat from the light source in the densitometer may cause them to deteriorate.
- Do not operate the densitometer when the environmental temperature falls below 10° C or rises above 40° C.
- Do not operate the densitometer when the environmental humidity falls below 20% or rises above 85%.

1.2 Introduction

The Model GS-700 Imaging Densitometer is a high performance, imaging densitometer that converts transparent and opaque electophoretic samples into digital data for analysis by the Molecular Analyst software operating with either Microsoft Windows® or Macintosh® computer systems.

The Model GS-700 Imaging Densitometer can scan reflectance samples up to 21 cm by 35.5 cm in the resolution range of 25 to 1200 dots per inch (dpi) and transmittance samples up to 20 cm by 25.5 cm in the resolution range of 25 to 1200 dots per inch (dpi).

The Model GS-700 Imaging Densitometer is supported by the Molecular Analyst™/PC and Molecular Analyst™/Mac Image Analysis Software, which provide versatile image display, optimization, and quantitation features.

The main features of the Model GS-700 Imaging Densitometer include:

- Adjustable resolution from 25 to 1200 dpi;
- Reflectance and transmittance modes;
- Stationary platen for higher reliability;
- Variable wavelength lamp/filter architecture for improved color discrimination;
- Analog to digital conversion at full 12-bit accuracy.

For a step-by-step guide to scanning and analyzing the resulting images, please refer to your Molecular Analyst software program user manual.

Section 2
Setting Up

2.1 Checking the Contents

Your Model GS-700 Imaging Densitometer System should arrive complete with one of each of the following:

- GS-700 Imaging Densitometer
- Power Cord
- Molecular Analyst™ Image Analysis Software
- 25/50-pin SCSI Cable (Macintosh)
- 50/50-pin SCSI Cable (PC)
- External SCSI Terminator
- Transparency Frame
- Diffuser Plate
- Reflectance Positioning Plate
- GS-700 Imaging Densitometer Manual
- Molecular Analyst Software Manual
- Warranty Card (please complete and return to Bio-Rad)
- Software Registration Card (please complete and return to Bio-Rad)

These components may arrive in several boxes. If your system is missing any of these items, immediately contact your local Bio-Rad office for a prompt replacement. Please retain all packaging material. Additional charges will be assessed if packaging is not available for instrument warranty service shipping.

Caution: Use care when unpacking your equipment. The equipment is delicate and damage can occur if it is dropped or otherwise mistreated.

2.2 Unpacking the Contents

Unpacking the Model GS-700 Imaging Densitometer shipping container.

1. Carefully lift the Model GS-700 Imaging Densitometer out of the shipping carton and place it on a stable, dry, flat surface. The chassis of the Model GS-700 Imaging Densitometer attempts to conform to any uneven surface. This may result in a poor-quality scan.

Warning: The Model GS-700 Imaging Densitometer weighs approximately 35.5 lbs. (16.1 kg). To avoid personal injury, two people may be needed to lift the densitometer out of the shipping container.

2. Remove the plastic wrapping and the packing materials from the densitometer.

3. The Model GS-700 Imaging Densitometer should be located close to the computer system to which it will be connected. Select a flat, dry surface that provides adequate ventilation on all sides to prevent overheating.

The Model GS-700 Imaging Densitometer can operate with 90V AC to 264V AC. Check the power cord to insure that it is suitable for your area.

Caution: The power cord plug provided has a third grounding pin. This plug fits in a grounding-type outlet only. This is a safety feature. If you are unable to plug the power cord into a wall outlet, locate another outlet that can accommodate the plug or contact your electrician to change the outlet. Do not change the plug.

2.3 Unlocking the Densitometer and the Transparency Module

The optical assemblies of the densitometer module (located in the base of the unit) and the transparency module (located in the lid) are held in place during shipment by a locking screw which must be loosened before the densitometer can be operated.

1. Turn the densitometer on its side and partially open the transparency module. If necessary, two people may be needed to stabilize the instrument while on its side.

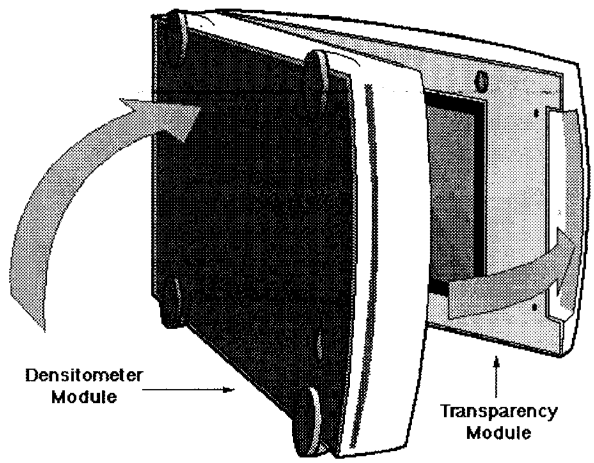

Fig. 2.1 Turning the densitometer on its side and opening the transparency module.

2. Unlock the locking screws by rotating counter-clockwise with a small coin (Figure 2.2). The screw pushes out a small distance, nearly even to the bottom of the densitometer and the transparency module.

Note: Leave the locking screws in place so that they can be relocked if you have to move the densitometer over long distances.

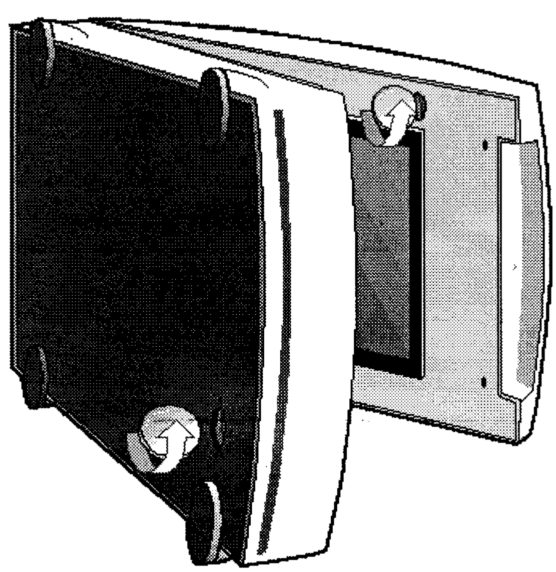

Fig. 2.2 Unlocking the densitometer and the transparency module.

3. Move the transparency module towards the body and turn the densitometer upright (Figure 2.3).

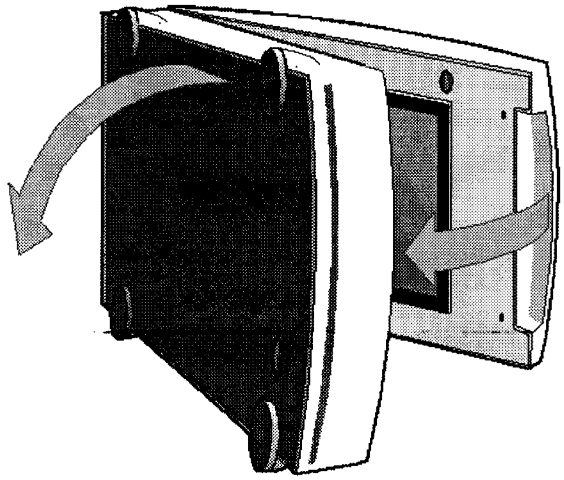

Fig. 2.3 Moving the transparency module and turning the densitometer upright.

4. Place the densitometer on a flat surface and do not move it suddenly. Sudden movements can cause the optical assembly to slide to the front or the back of the densitometer with enough force to damage the densitometer.

Warning: Always lower the transparency module slowly and release it only when it lies flat on the body of the densitometer.

2.4 Locking the Densitometer and the Transparency Module

Before transporting the densitometer over long distances, relock the locking screw. This will protect the optical assemblies of the densitometer and the transparency module from possible damage.

1. Turn on the densitometer. The Power indicator should light and the Scanner and Transparency Module indicators should blink. These indicator lights are located on the front right side of the densitometer. The optical assembly of the densitometer moves to its home position.

2. Wait until the Scanner indicator and the Transparency Module indicator stop blinking.

3. Turn off the densitometer and detach the power cord.

4. Turn the densitometer on its side and open the transparency module (Figure 2.4).

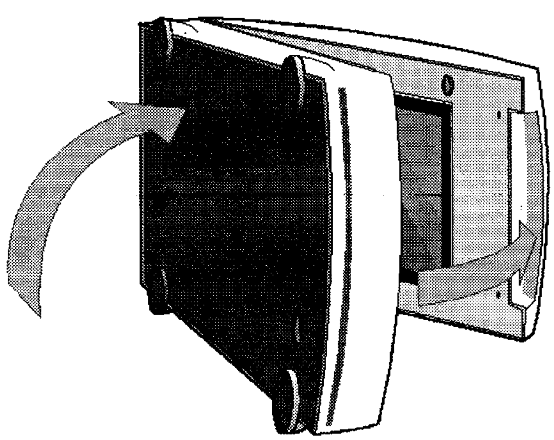

Fig. 2.4 Turning the densitometer on its side and opening the transparency module.

5. Lock the locking screws with a small coin by rotating clockwise (Figure 2.5). Push the locking screw in and tighten it.

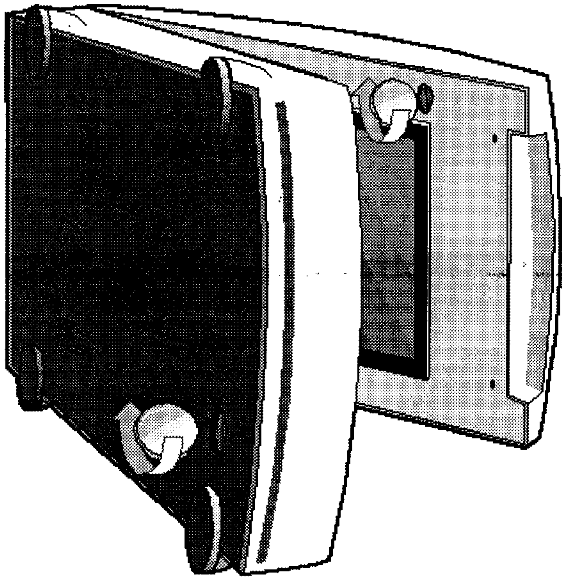

Fig. 2.5 Locking the densitometer and the transparency module.

6. Move the transparency module towards the body and turn the densitometer upright (Figure 2.6).

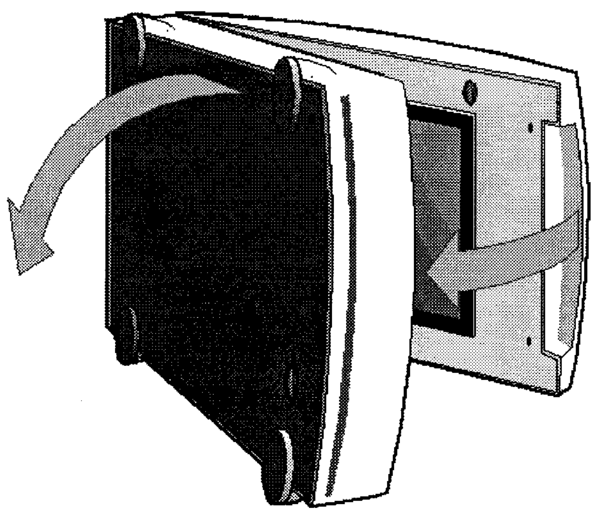

Fig. 2.6 Moving the transparency module and turning the densitometer upright.

7. Repack the densitometer with the original packing materials to protect it.

2.5 Taking a Closer Look

Now that you have the Model GS-700 Imaging Densitometer out of the box, take a closer look to familiarize yourself with the parts. Figures 2.7 and 2.8 illustrate the locations of the different parts of your Model GS-700 Imaging Densitometer.

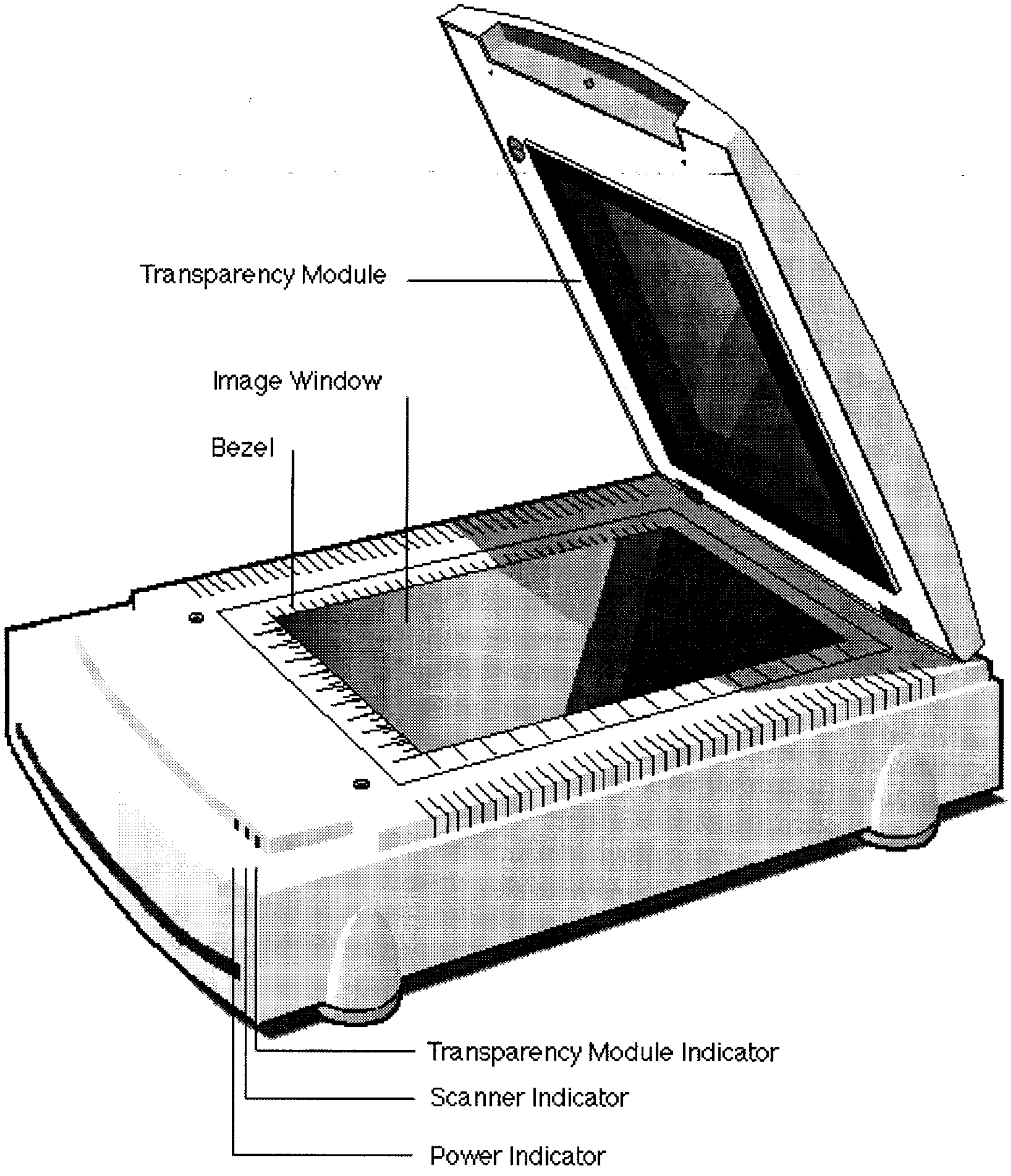

Fig. 2.7 The top and right side of the Model GS-700 Imaging Densitometer.

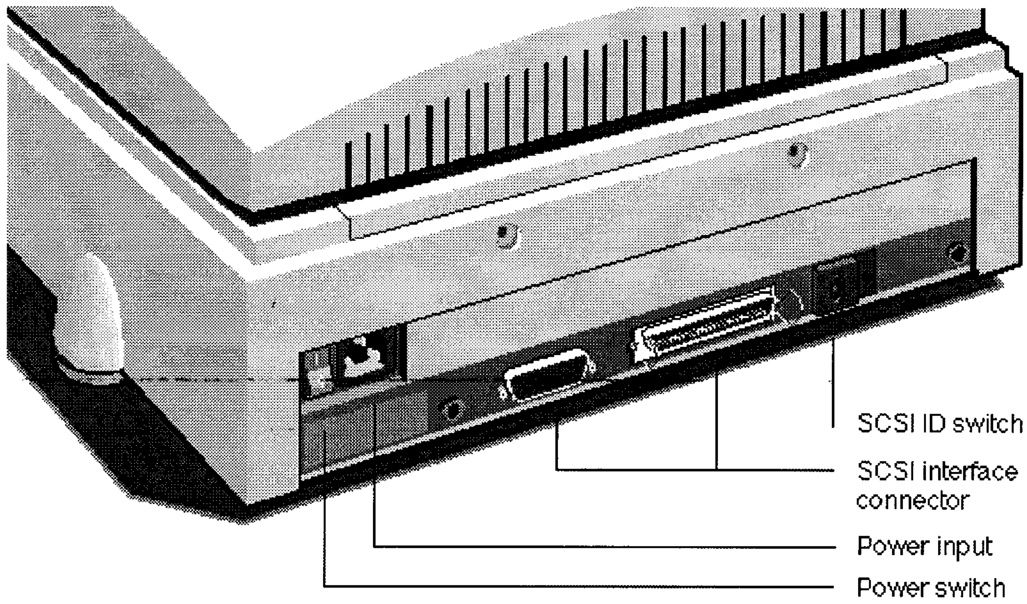

Fig. 2.8 The back of the Model GS-700 Imaging Densitometer.

2.6 Testing the Densitometer

You are now ready to perform a power-on test to check if the densitometer is operating correctly.

1. Check that you have removed the shipping restraints from the densitometer and from the transparency module.

2. Connect the power cord to the densitometer. Make sure that you are using the correct power cord for the voltage in your area. The Model GS-700 Imaging Densitometer automatically adjusts to any AC electrical outlet from 90V to 264V.

3. Turn on the densitometer with the power switch. The Power indicator lights up. The Scanner indicator and the Transparency Module indicator flash briefly. The scanning lamp turns on and should stay on while the densitometer is powered up. All three indicator lights will stay on while the densitometer is powered up.

The densitometer performs a self-test for approximately 35 seconds after which the Scanner indicator, the Transparency Module indicator and the scanning lamp stay on. If a problem occurs during the self-test, refer to Section 6, Troubleshooting.

Section 3
Installation

3.1 About SCSI Devices

The Model GS-700 Imaging Densitometer is a Small Computer System Interface (SCSI) device. It communicates with your computer by using the SCSI-2 standard. The SCSI communication standard allows you to connect more than one peripheral device to the same port of your computer in chain fashion.

A unique SCSI ID number is assigned to each device in the SCSI chain enabling your computer to identify the device with which it is communicating and the priority of each device.

Warning: If two SCSI devices have the same ID number, your system will not work properly and you may damage your SCSI devices (see Section 3.2).

A SCSI chain requires an electronic component called a 'terminator' which absorbs old signals traveling along the cables and keeps the path open for new signals. The chain should never have more than two terminators, one at each end. It is important to remember that using too many or too few terminators may damage your SCSI devices. Some SCSI devices have built-in terminators and must therefore be placed at the beginning or end of your SCSI chain.

Note: The Model GS-700 Imaging Densitometer does not have a built-in terminator.

3.2 Changing the SCSI ID Number

Before connecting the densitometer to your computer, you must be aware of the SCSI ID numbers already assigned. The Model GS-700 Imaging Densitometer is preset to ID2 and only needs to be changed if there is already another device in your chain set to the same number.

1. Make sure your densitometer is turned off and is disconnected from your computer.

2. Decide on an unassigned SCSI ID number.

3. Insert a pencil into one of the small holes above or below the SCSI ID number at the rear of the densitometer (Figure 3.1).

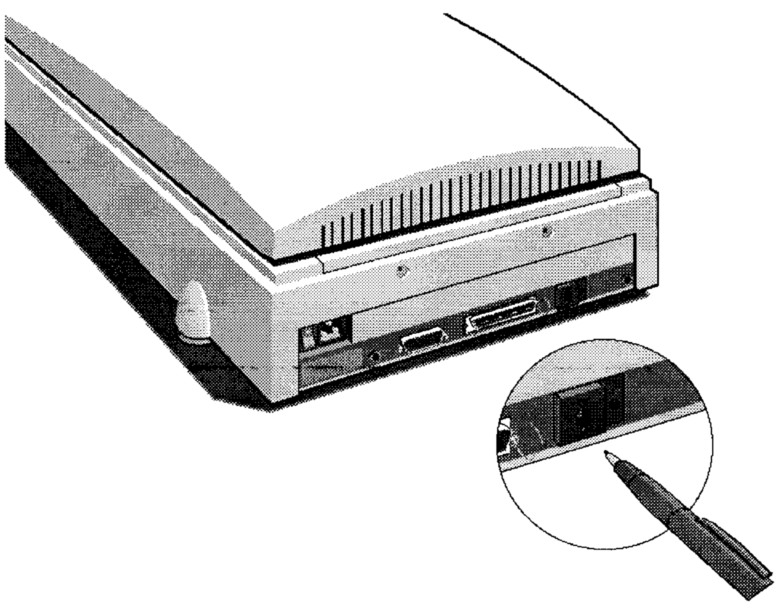

Fig. 3.1 Setting the SCSI ID number.

4. Push with your pencil above or below the SCSI number until you see the number you want. Push above the SCSI number to decrease the number, push below the SCSI number to increase the number.

3.3 Installation for the Macintosh

This section describes setting up the Model GS-700 Imaging Densitometer with your Macintosh computer. You must first choose and set a SCSI ID number, then connect the densitometer to your Macintosh, and lastly test the connection.

Choosing a SCSI ID Number

1. Determine which SCSI ID numbers are already assigned and which numbers are free. Your Macintosh always occupies ID7 and its internal hard disk usually occupies ID0 or ID1. The Model GS-700 Imaging Densitometer is preset to SCSI ID2.

2. If you need to change the SCSI ID number, see Section 3.2.

Connecting the Densitometer

Before you connect the densitometer to your Macintosh, make sure your Macintosh and all connected peripheral components are turned off.

If the Model GS-700 Imaging Densitometer is the only SCSI device to be connected to your Macintosh:

1. Connect the 25-pin end of the SCSI cable to the SCSI port on your Macintosh.
2. Place the terminator on the 50-pin SCSI port of your densitometer.
3. Connect the other side of the SCSI cable to the terminator that is placed on the 50-pin SCSI port of the densitometer (Figure 3.2).

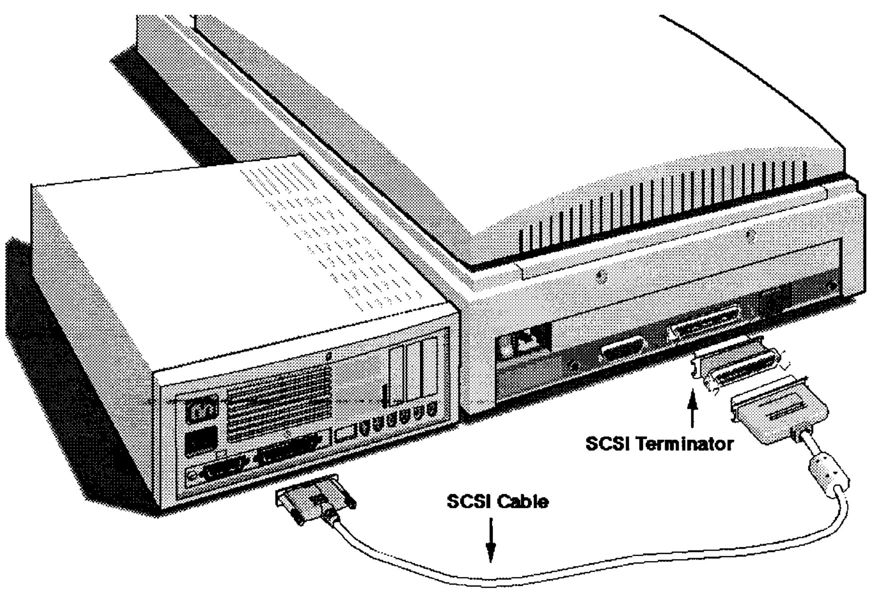

Fig. 3.2 Connecting the Model GS-700 to a Macintosh computer where the densitometer as the only SCSI device.

If the Model GS-700 Imaging Densitometer will be connected to your Macintosh with other SCSI devices:

1. Disconnect the two devices between which you want to place the densitometer. Leave the SCSI cable connected to the device that will be connected directly to the computer (cable A).

2. Connect the 50-pin end of the SCSI cable (cable B) that is already connected to another device to the 50-pin SCSI port of the densitometer (Figure 3.3).

3. Connect the 25-pin end of the new SCSI cable (25/50 pin cable) to the 25-pin SCSI port of the densitometer (cable C).

4. Connect the 50-pin end of this SCSI cable to the next device in the chain.

Warning: Make sure there are no more than two terminators in your SCSI chain. Some SCSI devices have built-in terminators and must therefore be placed at the beginning or end of your SCSI chain. Please check the documentation of each of your SCSI devices if you're not sure whether the device has a built-in terminator. The Model GS-700 Imaging Densitometer has no built-in terminator.

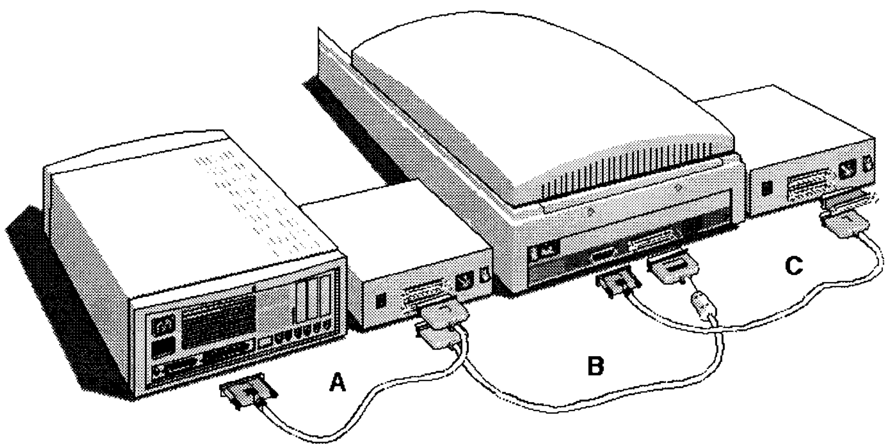

Fig. 3.3 Connecting the Model GS-700 to a Macintosh computer which is also connected to other SCSI devices.

If the Model GS-700 Imaging Densitometer will be installed at the end of your SCSI chain:

1. Remove the terminator from the last device in the SCSI chain.

2. Connect the 50-pin end of the SCSI cable to the SCSI port that has become available on this device.

3. Place the terminator on the 50-pin SCSI port of your densitometer.

4. Connect the other side of the SCSI cable to the terminator that is placed on the 50-pin SCSI port of the densitometer.

Warning: Make sure there are no more than two terminators in your SCSI chain. Some SCSI devices have built-in terminators and must therefore be placed at the beginning or end of your SCSI chain. Please check the documentation of each of your SCSI devices if you're not sure whether the device has a built-in terminator. The Model GS-700 Imaging Densitometer has no built-in terminator.

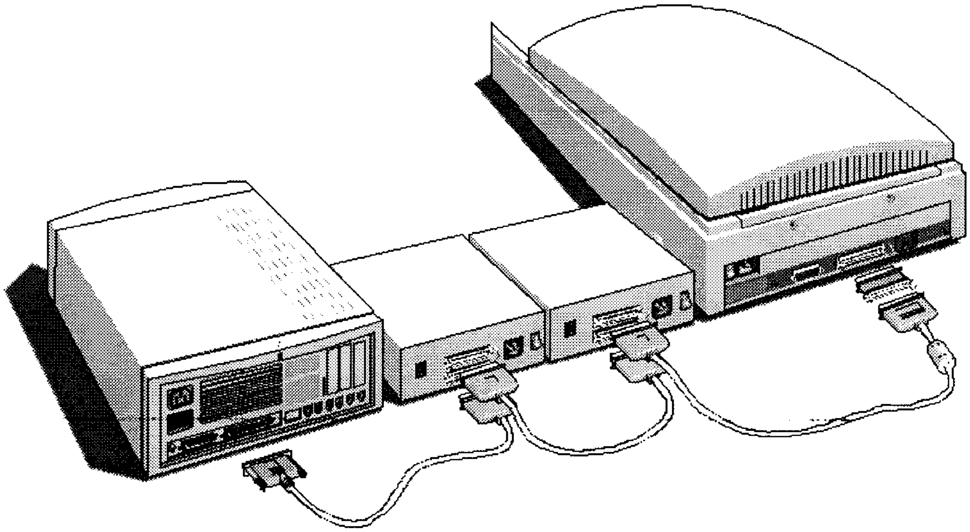

Fig. 3.4 Connecting the Model GS-700 to a Macintosh computer where the densitometer is at the end of the SCSI chain.

Testing the Connection

You are now ready to perform a test to check if the densitometer is correctly connected to your Macintosh.

1. Connect the power cord to the Model GS-700 Imaging Densitometer. The densitometer automatically adjusts to any AC electrical outlet rated from 90V to 264V.

2. Check that the SCSI cable(s) are properly connected.

3. Turn on the densitometer. After a self-test the Scanner and the Transparency Module indicator lights turn on.

4. Turn on any other SCSI devices you may have attached.

5. Turn on your Macintosh.

6. Install the Molecular Analyst software (see your software manual for instructions). Open the Molecular Analyst program. Under the FILE menu heading select ACQUIRE. Choose the densitometer imaging device. You should see the image capturing window displayed.

3.4 Installation for the PC

This section describes setting up the Model GS-700 Imaging Densitometer with your PC computer. You must first choose and set a SCSI ID number, then connect the densitometer to your PC, and lastly test the connection.

SCSI Adapter Card

The Model GS-700 Imaging Densitometer requires a SCSI adapter card to work with your PC-AT, 486, Pentium™, or compatible computer. The densitometer driver software supplied with the densitometer supports the Adaptec AHA 1540 and 1542 adapter cards.

Please check the following documentation:
- The Read Me file on the Molecular Analyst software disk for up-to-date information.
- The documentation supplied with your adapter card for instructions on card installation.

Choosing a SCSI ID Number

Determine which SCSI ID numbers are already assigned and which numbers are free. If you need to change the SCSI ID number, see Section 3.2. The Model GS-700 Imaging Densitometer is preset to SCSI ID2.

Connecting the Densitometer

Before you connect the densitometer to your PC, make sure your PC and all connected peripheral components are turned off.

If the Model GS-700 Imaging Densitometer is the only SCSI device to be connected to your PC:

1. Connect one end of the SCSI cable to the SCSI port at the rear of your PC (Figure 3.5).
2. Connect the other end of the SCSI cable to the corresponding SCSI port of the densitometer.
3. Place the terminator on the remaining SCSI port of your densitometer.

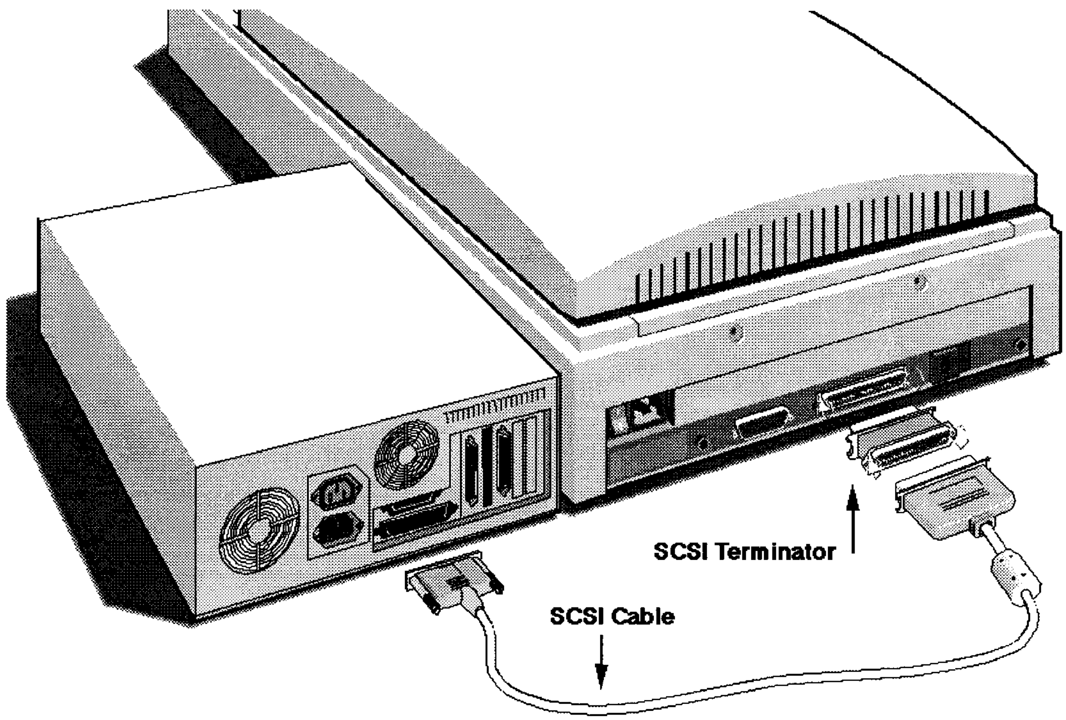

Fig. 3.5 Connecting the Model GS-700 to a PC computer where the Densitometer as the only SCSI device.

If the Model GS-700 Imaging Densitometer will be connected to your PC with other SCSI devices:

1. Disconnect the two devices between which you want to place the densitometer. Leave the SCSI cable connected to the device that will be connected directly to the computer (cable A).

2. Connect the 50-pin end of the SCSI cable (cable B) already connected to another device to the 50-pin SCSI port of the densitometer (Figure 3.6).

3. Connect the 25-pin end of the new SCSI cable (25/50-pin cable) to the 25-pin SCSI port of the densitometer (cable C).

4. Connect the 50-pin end of this SCSI cable (cable C) to the next device in the chain.

Warning: Make sure there are no more than two terminators in your SCSI chain. Some SCSI devices have built-in terminators and must therefore be placed at the beginning or end of your SCSI chain. Please check the documentation of each of your SCSI devices if you're not sure whether the device has a built-in terminator. The Model GS-700 Imaging Densitometer has no built-in terminator.

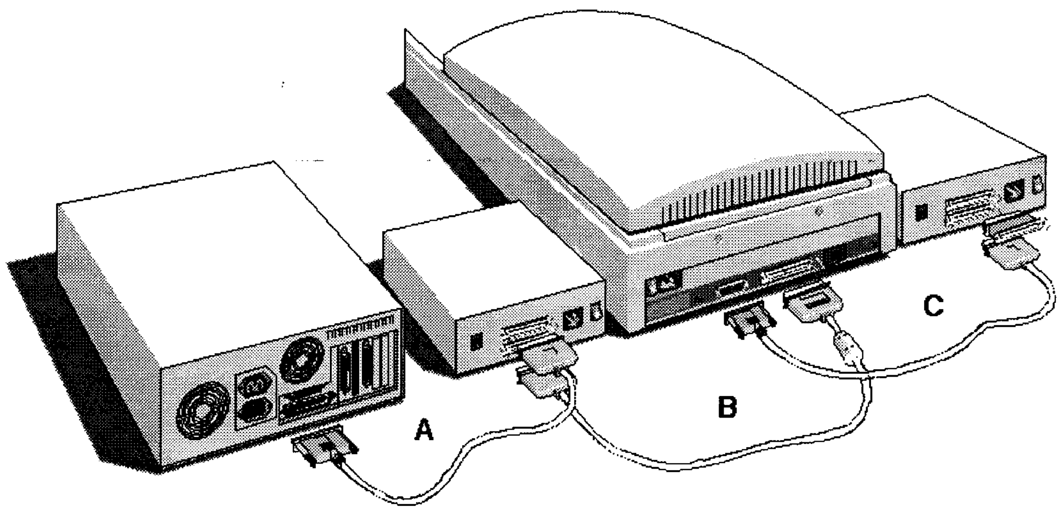

Fig. 3.6 Connecting the Model GS-700 to a PC computer which is also connected to other SCSI devices.

If the Model GS-700 Imaging Densitometer will be installed at the end of your SCSI chain:

1. Remove the terminator from the last device in the SCSI chain.

2. Connect the 50-pin end of the SCSI cable to the SCSI port that has become available on this device.

3. Place the terminator on the 50-pin SCSI port of your densitometer.

4. Connect the other side of the SCSI cable to the terminator that is placed on the 50-pin SCSI port of the densitometer.

Warning: Make sure there are no more than two terminators in your SCSI chain. Some SCSI devices have built-in terminators and must therefore be placed at the beginning or end of your SCSI chain. Please check the documentation of each of your SCSI devices if you're not sure whether the device has a built-in terminator. The Model GS-700 has no built-in terminator.

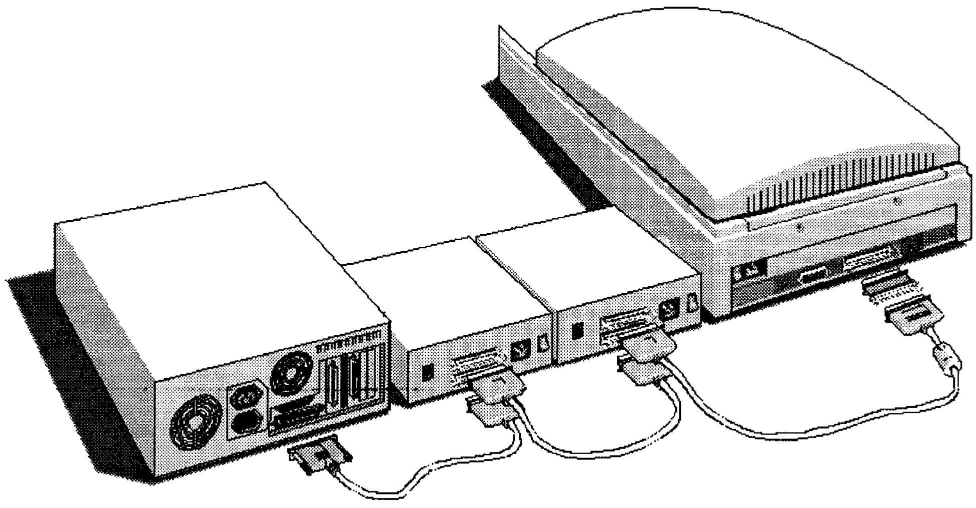

Fig. 3.7 Connecting the Model GS-700 to a PC computer where the Densitometer is at the end of the SCSI chain.

Testing the connection

You are now ready to perform a test to check if the densitometer is correctly connected to your PC.

1. Connect the power cord to the Model GS-700 Imaging Densitometer. The densitometer automatically adjusts to any AC electrical outlet rated from 90V to 264V.

2. Check that the SCSI cable(s) are properly connected.

3. Turn on the densitometer. After a self-test the Scanner and the Transparency Module indicator lights turn on.

4. Turn on any other SCSI devices you may have attached.

5. Turn on your PC.

6. As it boots up, your PC performs a series of tests to verify the correct system configuration. In case of problems, refer to Section 6 'Troubleshooting'. Install the Molecular Analyst software (see your software manual for instructions). Open the Molecular Analyst program. Make sure the Model GS-700 Imaging Densitometer is selected as the current scanner in the PREFERENCES window under the FILE menu heading. Under the FILE menu heading select NEW. You should see the image capturing window displayed.

Section 4
Operation

4.1 Overview of Operational Components

This chapter provides a summary of the Model GS-700 Imaging Densitometer components used during operation and explains the startup procedure. Several densitometer components are used during reflectance and transmittance operation.

Transparency Module
The Transparency Module (Figure 2.7) is secured to the Model GS-700 Imaging Densitometer by two elongated hinges at the rear of the densitometer. This Transparency Module should be lowered onto the densitometer base for reflectance and transmittance scanning. The thick border gasket on the transparency module allows you to place samples up to 3.0 mm thick on the image window for scanning.

Warning: Always lower the transparency module slowly and release it only when it lies flat on the body of the densitometer.

Transparency Frame
When you scan a transparent sample, place the transparency frame (Figure 4.1) on the densitometer's image window with the calibration hole facing the front. Place your sample in the transparency frame. Do not cover the calibration hole or a calibration error will occur. If you do not use the transparency frame during a transparent scan, leave a gap of at least 2 inches (5.1 cm) between the left edge of the densitometer bezel and the top of the sample.

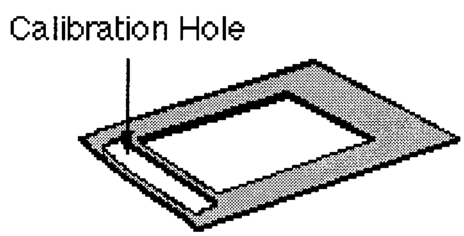

Fig. 4.1 The transparency frame.

Bezel
The bezel (Figure 4.2) is the raised frame around the image window. It has marks in inches and centimeters to help you correctly position the sample on the image window for scanning.

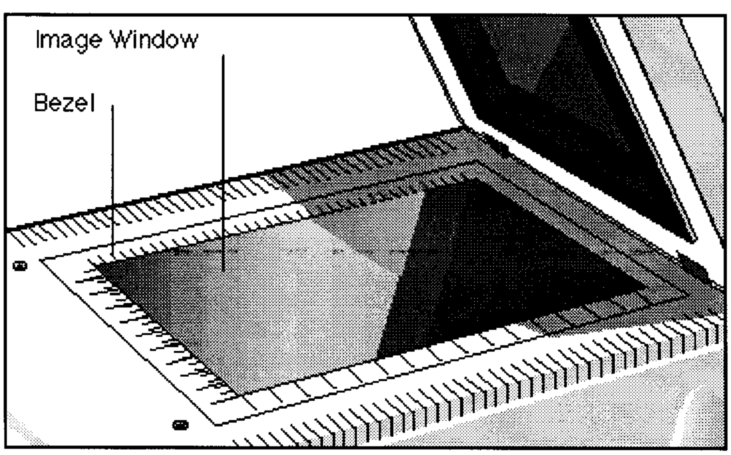

Fig. 4.2 The image window and bezel on the Model GS-700 Imaging Densitometer.

Image Window
The image window (Figure 4.2) is the area where the sample to be scanned is positioned. Light reflected from the sample is focused through a lens onto a charge coupled device (CCD) array that captures the image.

Caution: The image window is glass. To avoid breakage, do not press firmly or place anything other than a sample for scanning on the image window.

Indicator Lights
The indicator lights (Figure 2.7) are located on the front right side of the Model GS-700 Imaging Densitometer cabinet. This panel consists of three lights that indicate the status of the unit.

Light	Description
Power Indicator	Is lit when power switch is in the on position (green light).
Scanner Indicator	Is lit during the power-up cycle and when the densitometer is ready to accept scanning commands (orange light). This light will blink a few seconds before and after a reflectance scan.
Transparency Module Indicator	Is lit during the power-up cycle and when the densitometer is ready to accept scanning commands (orange light). This light will blink a few seconds before and after a transparency scan.

Diffuser Plate
The diffuser plate (8 x 10 inches - transparent gray plate) insures that light is evenly dispersed over the entire sample to provide you with a quality image during a transparency scan. The diffuser plate is not needed for scanning X-ray film, but is needed with transparent wet or dried gels.

When using the diffuser plate with wet gels care must be taken to eliminate air bubbles from getting trapped between the gels and the plate. First, add extra water to the top of the gel. Second, place one end of the diffuser plate on the image window of the densitometer. Then, slowly lower the diffuser plate over the gel while allowing air to escape through the other end.

When using the diffuser plate, make sure that it does not cover the calibration hole in the transparency frame.

Reflectance Positioning Plate
The reflectance positioning plate (8.5 x 13 inches - white Plexiglas plate) can be used during a reflectance scan. This plate helps to flatten out wrinkled samples and keep them close to the image window for good resolution.

4.2 Start-Up

The Model GS-700 Imaging Densitometer on/off control is a rocker switch located on the back of the unit to the right of the power cord receptacle. When the top side of the rocker is depressed, the switch is on. To turn the power off, depress the bottom side of the rocker.

1. Turn on the densitometer. The Power indicator lights up. The densitometer performs a self-test that lasts approximately 35 seconds in which the Scanner indicator and the Transparency Module indicator flash briefly. The scanning lamp turns on and all three indicator lights will stay on while the densitometer is powered up.

2. For optimum performance, allow the densitometer to warm up for five minutes after power-up.

3. The Model GS-700 Imaging Densitometer is now ready for use.

4.3 Reflective Scanning

To scan a reflective sample:

1. Raise the Transparency Module.
2. Place the sample to be scanned face-down on the image window.
3. Guide the top corner of the sample to the lower left (0,0) corner of the image window.
4. The reflectance positioning plate can be used to flatten out wrinkled samples.
5. Lower the Transparency Module.

Warning: Always lower the transparency module slowly and release it only when it lies flat on the body of the densitometer.

6. For more information about scanning operations and options, refer to the Molecular Analyst software manual.

Keep the image window clean and free of oil and fingerprints. See Section 5, Maintenance, for cleaning procedures.

4.4 Transmittance Scanning

To scan a transparent sample:

1. Raise the Transparency Module.
2. Place the Transparency frame on the image window with the calibration hole facing the front of the densitometer.
3. Place the sample face down on the densitometer's image window.
4. Guide the top corner of the sample to the lower left (0,0) corner of the image window.
5. The diffuser plate can be used for wet or dried gels.
6. Lower the Transparency module.

Warning: Always lower the transparency module slowly and release it only when it lies flat on the body of the densitometer.

7. For more information about scanning operations and options, refer to the Molecular Analyst software manual.

Keep the image window clean and free of oil and fingerprints. See Section 5, Maintenance, for cleaning procedures.

Section 5
Maintenance

The Model GS-700 Imaging Densitometer requires easy periodic cleaning for continued optimum performance.

Warning: To prevent personal injury, always turn the power switch off and unplug the power cord from the wall outlet before performing any maintenance on the Model GS-700 Imaging Densitometer.

Caution: Do not use ammonia-based cleaning products to clean the Model GS-700 Imaging Densitometer. Do not disassemble the Model GS-700 Imaging Densitometer or lubricate any parts.

5.1 Cleaning the Image Window Glass

To clean the glass:

1. Turn off the power switch.
2. Unplug the power cord from the wall outlet.
3. Use a slightly damp cloth with a small amount of mild detergent or alcohol to wipe the glass.
4. Dry the glass with a clean, lint-free cloth.

Caution: To avoid breakage, do not rub heavily on the glass. Do not spray liquids directly onto the image window and do not use a very wet cloth to clean the glass. Excess liquid can seep underneath the glass and possibly damage the internal components. If spots remain on the glass, carefully use a razor blade to remove them.

5.2 Cleaning the Cabinet Exterior

To clean the cabinet exterior:

1. Turn off the power switch.
2. Unplug the power cord from the wall outlet.
3. Use a slightly damp cloth with a small amount of mild detergent or alcohol to wipe the glass.
4. Dry the cabinet with a clean, lint-free cloth.

Caution: If the cabinet becomes stained and a damp cloth does not remove the stain, a mild cleaning solution can be used.

5.3 Replacing the Fluorescent Lamps

Replacing the Lamp on the Densitometer

To replace the lamp of the Model GS-700 Imaging Densitometer, only use the 8W fluorescent light specified by Bio-Rad (Part Number 100-1450).

1. Turn off the power switch.
2. Disconnect the power plug.

3. Wait for at least 15 minutes for the temperature to drop sufficiently before attempting replacement. The lamp housing and the lamp can be extremely hot.

4. Loosen the 2 screws of the main board at the back of the densitometer (Figure 5.1).

5. Pull the board out of the densitometer so that you can remove the connector.

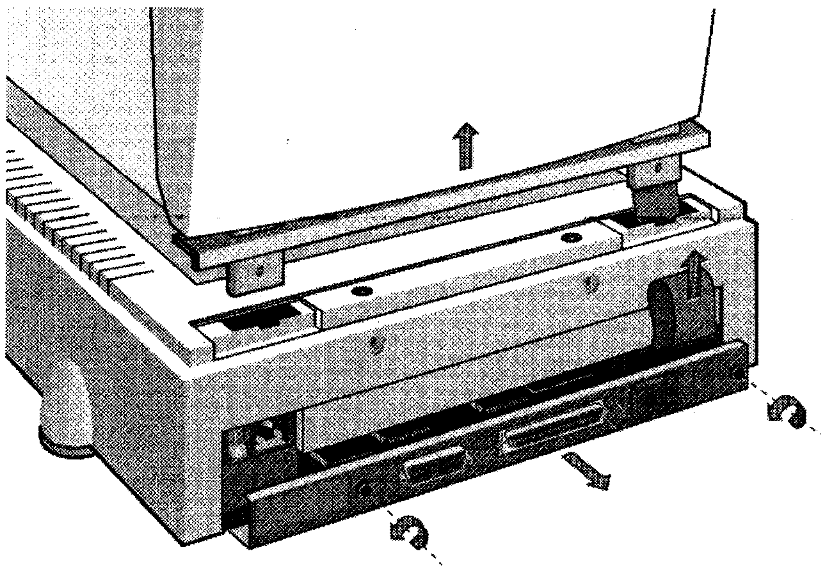

Fig. 5.1 Removing the connector.

6. Raise the transparency module and carefully remove it from the densitometer.

7. Remove the 4 top screws as indicated in Figure 5.2.

8. Remove the housing of the densitometer as shown in Figure 5.2.

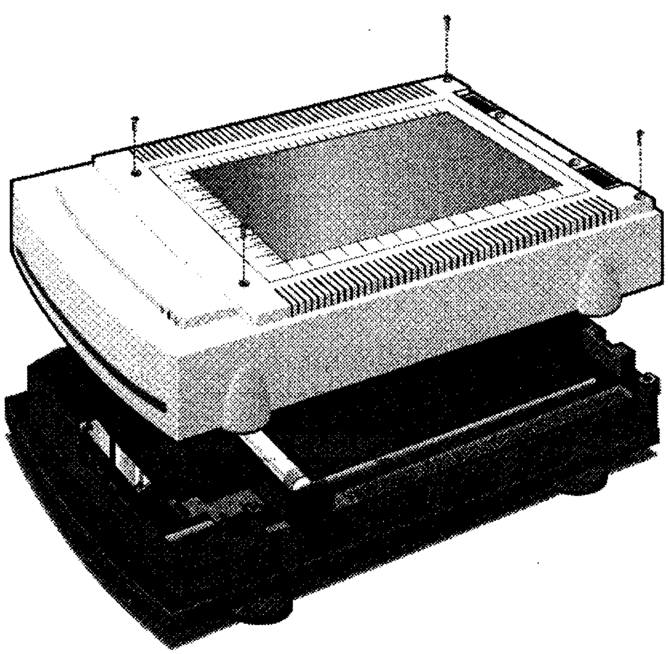

Fig. 5.2 Removing the screws and the housing of the densitometer.

9. Push the densitometer's optical assembly backward until you can reach the lamp.

10. Lift the lamp from the sockets as shown in Figure 5.3.

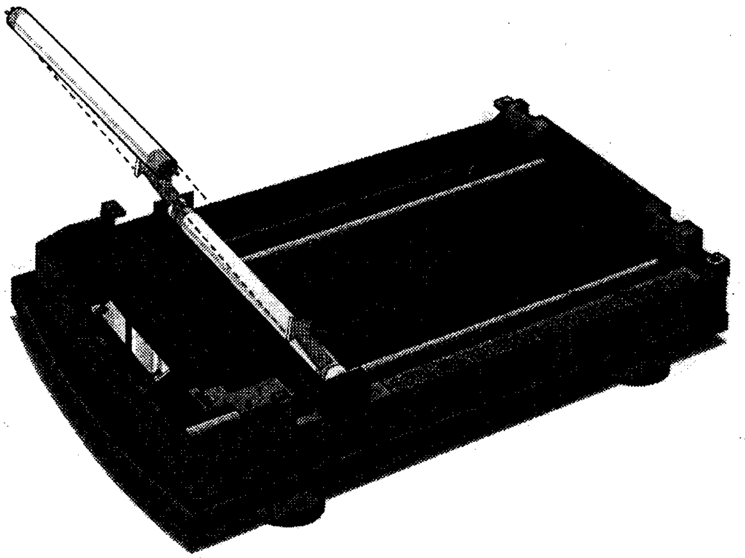

Fig. 5.3 Removing the lamp.

11. Place the new lamp in the sockets.

12. Replace the densitometer housing and its 4 screws.

13. Replace the transparency module.

14. Reconnect the connector.

15. Replace the main board and tighten the 2 screws.

Replacing the Lamp on the Transparency Module

To replace the lamp of the Transparency Module, only use the 8W fluorescent light specified by Bio-Rad (Part Number 100-1450).

1. Turn off the power switch.

2. Disconnect the power plug.

3. Wait for at least 15 minutes for the temperature to drop sufficiently before attempting replacement. The lamp housing and the lamp can be extremely hot.

4. Remove the transparency module from the densitometer and carefully place it upside down on a table.

5. Remove the 11 screws as indicated in Figure 5.4.

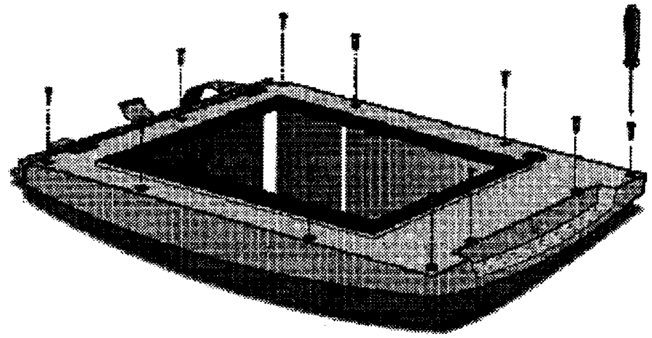

Fig. 5.4 Removing the screws.

6. Turn the transparency module over again (right-side up).

7. Remove the housing of the transparency module (Figure 5.5).

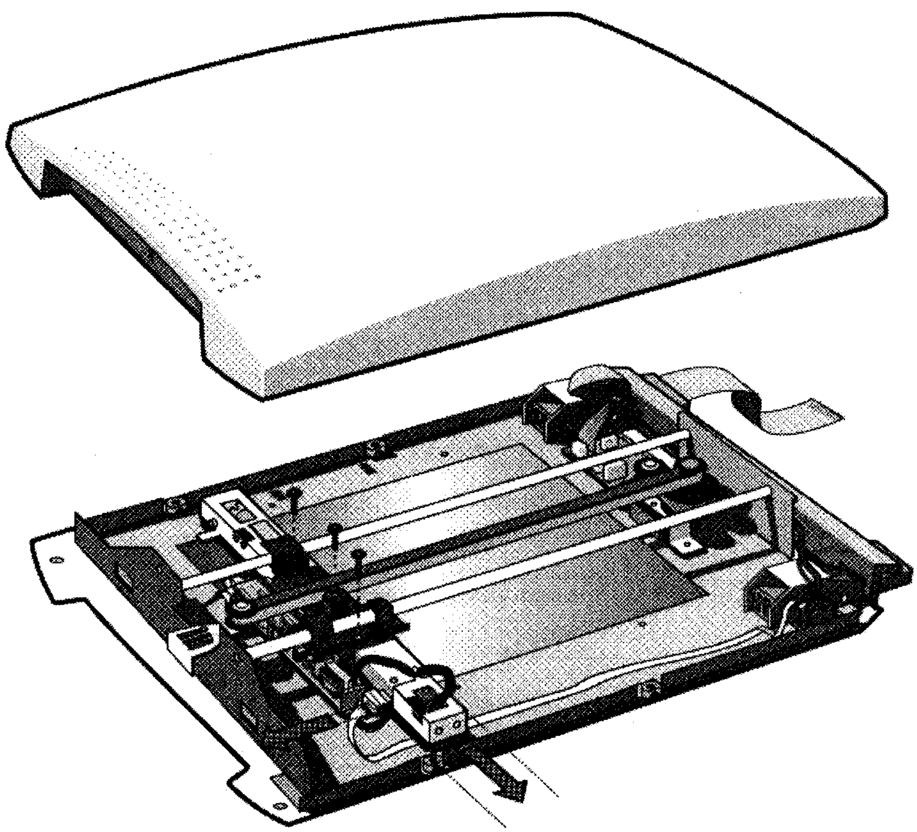

Fig. 5.5 Removing the housing of the transparency module.

8. Push the transparency module's optical assembly backward until you can reach the screws of the lamp cover.

9. Remove the top three screws on the lamp assembly (Figure 5.5).

10. Remove the flexible cable and connector (Figure 5.5).

11. Slowly turn the lamp assembly 90 degrees and carefully remove from the transparency module. Be careful not to damage the glass plate of the transparency module.

12. Remove the two screws from the invertor board as indicated in Figure 5.6 (A).

13. Remove the side three screws on the lamp cover as indicated in Figure 5.6 (B).

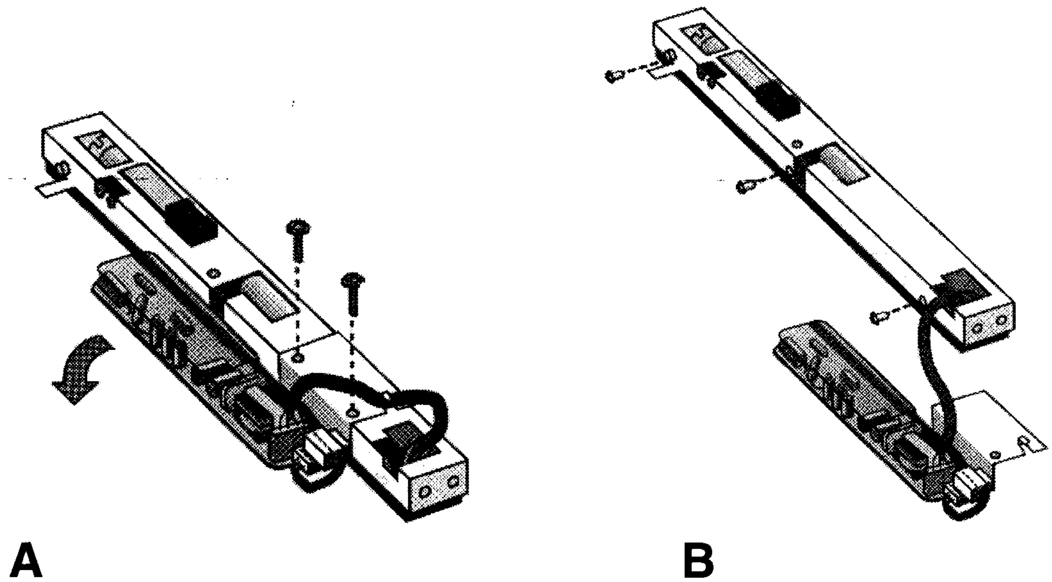

Fig. 5.6 Removing the screws from the lamp cover.

14. Remove the lamp cover and lift the lamp from the sockets (Figure 5.7).

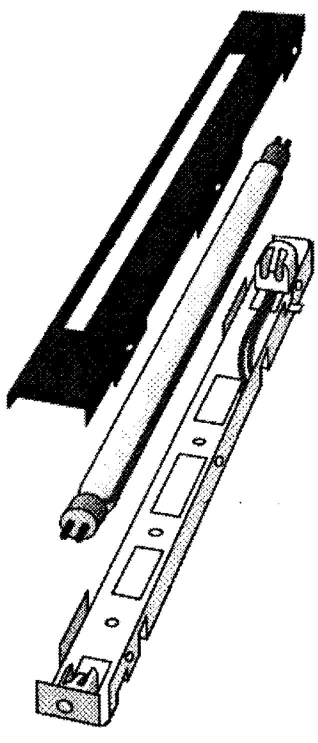

Fig. 5.7 Removing the lamp from the lamp cover.

15. Fit the new lamp into the sockets.

16. Reinstall the lamp cover and its three screws.

17. Reinstall the invertor board bracket and its two screws.

18. Reinstall the lamp assembly in its original place and reinstall the three screws that hold the lamp in place.

19. Reinstall the flexible cable and connector.

20. Reinstall the transparency module's housing and its eleven screws.

21. Reinstall the transparency module on the densitometer.

Section 6
Troubleshooting

6.1 Troubleshooting

Though the Model GS-700 Imaging Densitometer is very reliable, problems can occur during operation. This section lists some problems and the recommended corrective actions. Most problems are easily remedied.

Electronic Failure
Call Bio-Rad Technical Service (1-800-424-6723).

Will not scan again, stays in home position, or grinds.
This type of condition is probably related to a mechanical failure somewhere inside the densitometer. A part may have come loose and is obstructing the path of the lamp assembly. Contact Bio-Rad Technical Service.

Both the Scanner indicator and the Transparency Module indicator on the operating panel fail to light up.

1. Check that you have removed the locking screws.
2. Check the scanning lamp in the densitometer and in the transparency module. If the lamp flickers, dims, or goes off, please contact Bio-Rad Technical Service.
3. If the weather is very cold, leave the densitometer on for several minutes to let it warm up. Turn the power off, then immediately on again. If both lights still go off, contact Bio-Rad Technical Service.

The Transparency Module lamp flickers, is dim, or fails to come on.

1. The scanning lamp is failing or has failed and needs to be changed. Refer to Section 5 on replacing the scanning lamp.
2. The lamp may have come loose. Readjust the lamp.

The Power indicator fails to light up.

1. Verify the power connection to the densitometer.
2. Check that the power switch is turned on.
3. If you have confirmed that there is power to the densitometer, it is likely that the densitometer fuse needs to be replaced. Contact Bio-Rad Technical Service.

The scanned image is marred by specks or horizontal lines.
Hardware failure. Contact Bio-Rad Technical Service.

The scanned image is showing random dark specks or vertical L bands.
Clean the image window glass. Refer to Section 5.1 on Maintenance.

The scanned image is marred by Moiré patterns
Moiré patterns are often seen when scanning a halftone. A halftone is a picture in which the gradations of light are simulated by dots of varying sizes or densities, produced by superimposing a screen over an image (e.g., pictures in newspapers and magazines). It is best to avoid scanning halftones because of the moiré, but if it is unavoidable, scan at a lower resolution.

The scanned image is too dark.

1. If your are scanning a transparent sample, make sure your are using the transparency selection.
2. If the problem persists, contact Bio-Rad Technical Service.

The scanned image is too light, or is washed out.
If you are using the transparency option, try using the diffuser plate.

The computer does not boot.
Your computer cannot find its hard disk due to a conflict with the SCSI ID numbers of the devices you have attached. Turn off the power to the densitometer and unplug the cord. Disconnect all SCSI devices (except the startup disk) and connect them one by one, beginning with the densitometer, to identify the device that causes the problem.

The Molecular Analyst software returns a "Densitometer not ready" or similar message.
1. Verify the power connection to the densitometer.
2. Check that the power switch is turned on.
3. Check the Installation procedure (Section 3), to see if you followed the instructions. Pay special attention to the setting of the SCSI ID number.
4. Verify the integrity of the terminators and the cables. If there is a problem with a cable or with the SCSI board of the densitometer, contact Bio-Rad Technical Service.
5. Disconnect all SCSI devices (except the startup disk) and connect them one by one, beginning with the densitometer, to identify the device that causes the problem.
6. Verify that densitometer imaging device is selected in the FILE menu of the Molecular Analyst software.

6.2 Technical Service

Bio-Rad provides complete technical support on all features of the GS-700 Imaging Densitometer system, including all hardware, software, or spare parts issues. If you have any problems or any questions on the use of the features, contact your local Bio-Rad office, or in the U.S. call 1-800-424-6723. In order to fully benefit from our technical service, please complete your registration card and return it to the address shown on the card.

Section 7
Technical Specifications

Color Separation	Single CCD sensor in conjunction with fluorescent lamps and three RGB filters
Scanning Resolution	Variable resolution from 25 to 1200 dpi
Gray Scales	12-bit A/D conversion
Effective Scanning Area	21 cm x 35.5 cm Reflective Mode (8.5 in. x 14 in.) 20 cm x 25.5 cm Transmittance Mode (8 in. x 10 in.)
Interface	SCSI interface - double connector
Transmission Speed	1 Mbyte/second
Lamp	8W daylight fluorescent light
Warm up time	35 seconds, 5 minute for stable operation
Voltage	Line voltage: 90 to 264 volts Frequency: 47 to 63 Hz Power: 40 watts maximum
Dimensions	Length: 60 cm (23.6 in.) Width: 41 cm (16.1 in.) Height: 20 cm (7.9 in.)
Weight	Approximately 16.1 kg (35.5 lb)
Environment	Operating temperature 10°C to 40° C (50° F to 104° F) Storage temperature -25° C to 55° C (-13° F to 131° F) Relative Humidity 20% to 95% Surrounding space 15 cm on each side

Section 8
Ordering Information

Description	Part Number
Diffuser Plate (1)	9200351
Reflectance Positioning Plate (1)	9200349
Transparency Frame (1)	4200164
Fluorescent Lamp (1)	1001450